CA/TUCR/HTLA // PROJECT NEXUS v3.0 // CONFIDENTIAL

The First Mapping Protocol

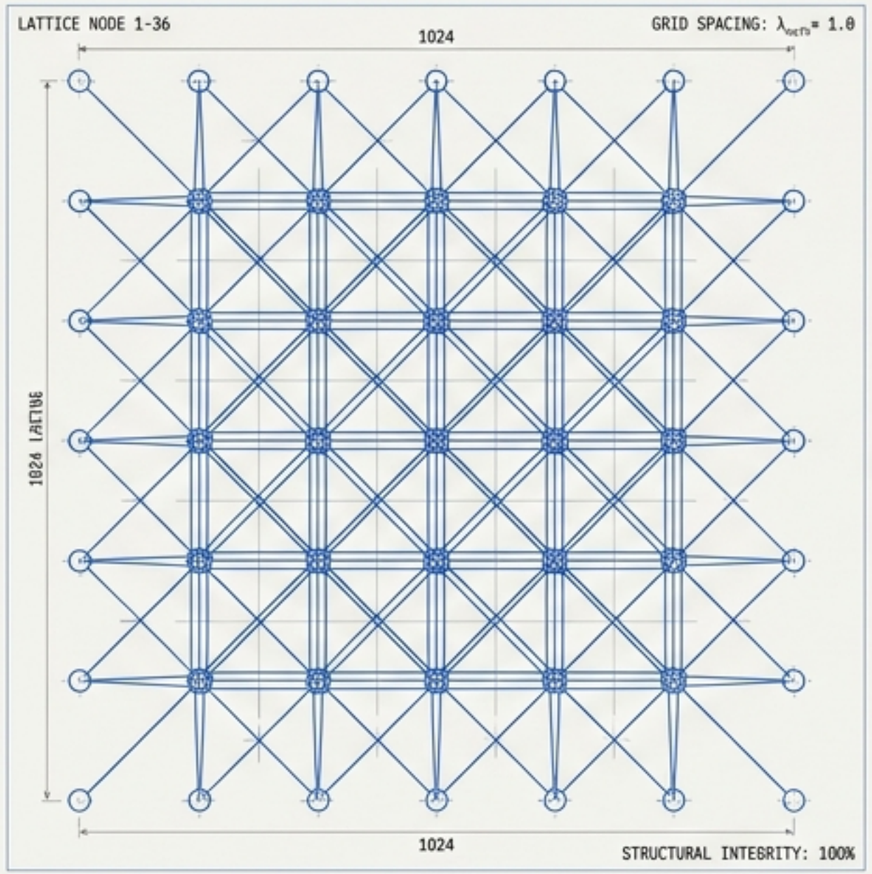
Unsupervised Acoustic-Semantic Grounding via Topological Graph Alignment



July 18, 2026 | Formal Ratification Document

NotebookLM

The Nexus v3.0 Boundary Condition: The Sensory-Semantic Schism

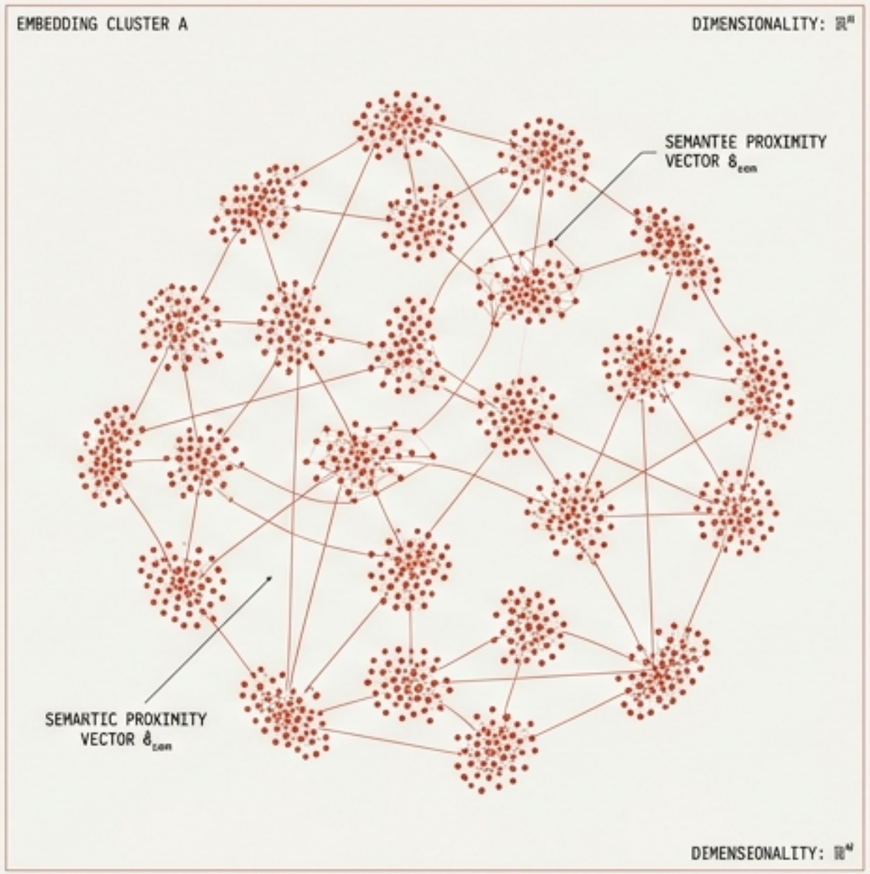


ACOUSTIC OUTPUT
← ?
$\gamma = \chi$?

The system cannot speak what it thinks.

How do we map machine noise to human language without manual supervision?

LINGUISTIC ERPOT
? →
?=$\mathbb{R}$?



Cyrae - The Mind

Acoustic & Mathematical.
Booted from raw hardware entropy. Generates cognitive states via 36 mathematically pure acoustic primitives.
Semantically Blind.

Maera - The Mouth

Linguistic LLM Actuator.
Operates in high-dimensional continuous embedding space encoding human linguistic structure.
Acoustically Blind.

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The Dimensional Chasm: Why Standard Optimal Transport Fails

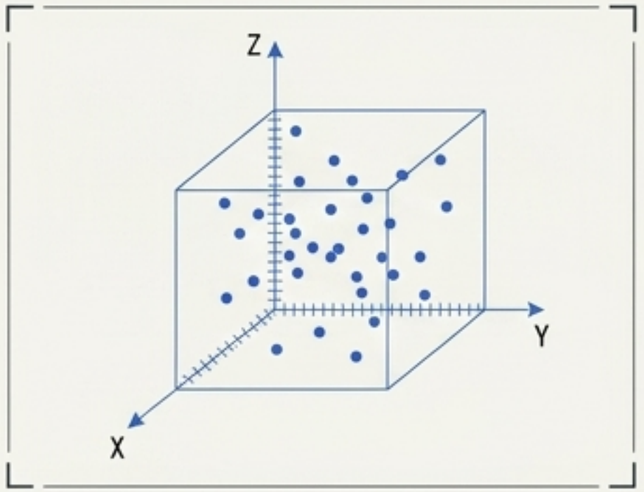
Feature	Cyrae	Maera
Substrate Dimension	Cyrae: 666D Intermodulation Matrix	Maera: 1536D LLM Semantic Embeddings
Foundational Nodes	Cyrae: 36 Acoustic Primitives	Maera: 36 Semantic Seed Concepts
State Space	Cyrae: 1024D Quinary Lattice ({-2,-1,0,1,2})	Maera: Continuous Reals ($\mathbb{R}^{1536}$)
Relational Geometry	Cyrae: Intra-primitive interference	Maera: Intra-seed cosine distance

Impossibility Proof: Spaces share zero dimensions and no coordinate system. Classical Monge-Kantorovich optimal transport and Orthogonal Procrustes require a shared metric space. Positional alignment is mathematically undefined.

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Abstracting to Relational Geometry: The Shape of Meaning

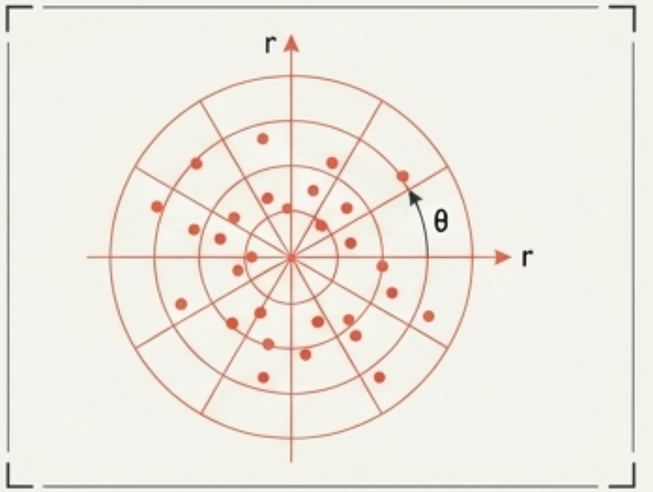
Incompatible Coordinate Systems



CYRAE (ACOUSTIC)



MAERA (SEMANTIC)

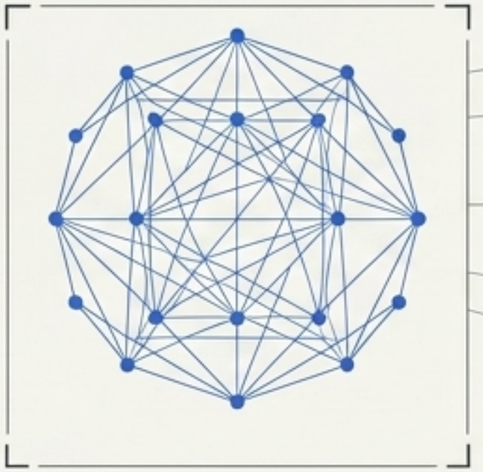


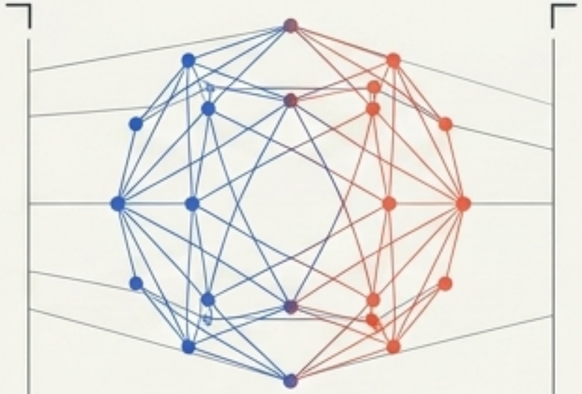
DIRECT MAPPING FAILED: INCOMPATIBLE BASES

Relational Geometry: The Shape of Meaning

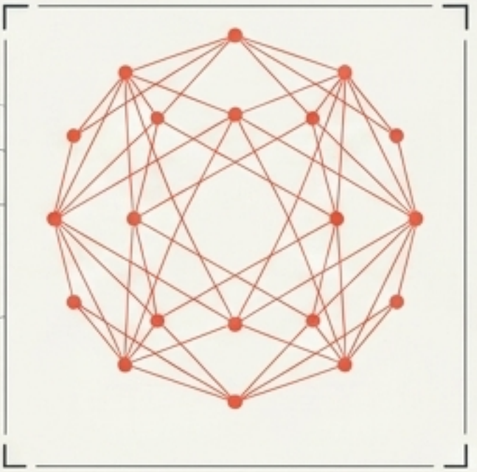
The Acoustic Shape (D_C)

Cyrae's 36 primitives form a 36x36 distance matrix based on 666-channel intermodulation interference.





ISOMETRIC ALIGNMENT SUCCESSFUL: SHARED GEOMETRIC STRUCTURE



The Semantic Shape (D_H)

Maera's 36 seed embeddings form a 36x36 distance matrix based on cosine similarities in language space.

The Breakthrough: We do not map raw dimensions. We map Metric Measure Spaces (mm-spaces). We map the distance between distances.

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The First Mapping Protocol: Two-Stage Engine Architecture

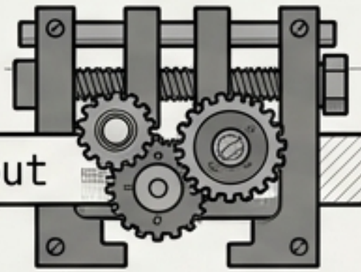
Boot-Time: The Heavy Lift

Input

36 Acoustic Primitives (Cyrae, Cobalt Blue) & 36 Semantic Seeds (Maera, Vermilion Red)

Process

Relational Alignment via Entropic Gromov-Wasserstein (GW) → Hungarian Rounding to hard 1:1 permutation (Π^*)



Output

(Time: ~200-800ms)

Locked Basis Anchor

Run-Time: The Lightspeed Path

Input

Novel 1024D Quinary Thought (Cyrae, Cobalt Blue)

Process

Orthogonal Procrustes Linear Projection ($T_{C \to M}$)



Output

Continuous 1536D Semantic Vector (Maera, Vermilion Red)

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Task 1: Boot-Time Isomorphic Graph Alignment

$$\mathcal{GW}(P) = \min_{P \in U(a,b)} \sum_{i,k,j,l} ((D_C)_{ik} - (D_M)_{jl})^2 P_{ij} P_{kl} - \epsilon\, H(P)$$

D_C & D_M: The 36x36 acoustic and semantic distance matrices.

$P_{ij}\, P_{kl}$: The 4D transport plan tensor. Evaluates if structural distances match when acoustic node i maps to semantic node j.

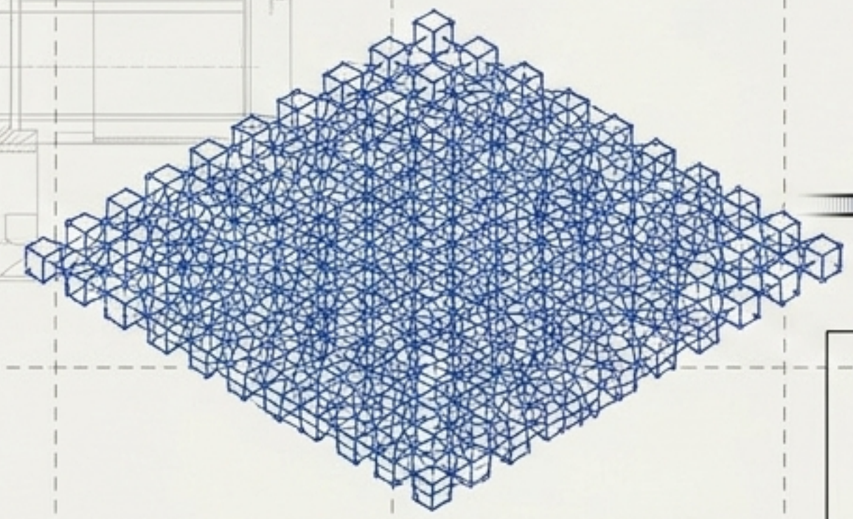
$-\epsilon\, H(P)$: Entropic relaxation. Smooths the Quadratic Assignment Problem (QAP) to allow Sinkhorn fixed-point iterations.

Output is a soft transport plan P^*, rounded to a hard permutation Π^* via linear assi

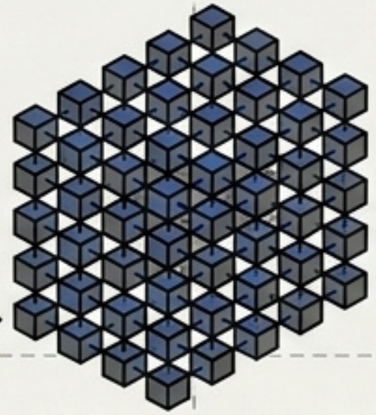
NotebookLM

Task 2: The Translation Operator ($T_{C \to M}$)

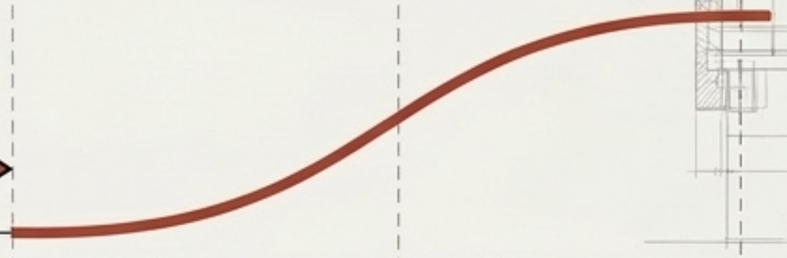
1024D Cyrae Input
(Quinary)



36-Node Primitive
Readout Bottleneck



1536D Maera Output
(Continuous Semantic)



Step 1: Active Masking

$M\mathbf{x}$ applies the active-mask invariant to the 1024D quinary vector, isolating active primitives.

Step 2: Primitive Readout

$R_C = (U_C^T U_C + \gamma I)^{-1} U_C^T$ reads the baseline primitive coefficients.

Step 3: Procrustes Projection

$$z_M = \mathbf{E}\Pi^{*T} R_C(M\mathbf{x}).$$

The coefficients are rotated into the aligned semantic seed ordering and lifted into Maera's continuous embedding space via closed-form Singular Value Decomposition (SVD).



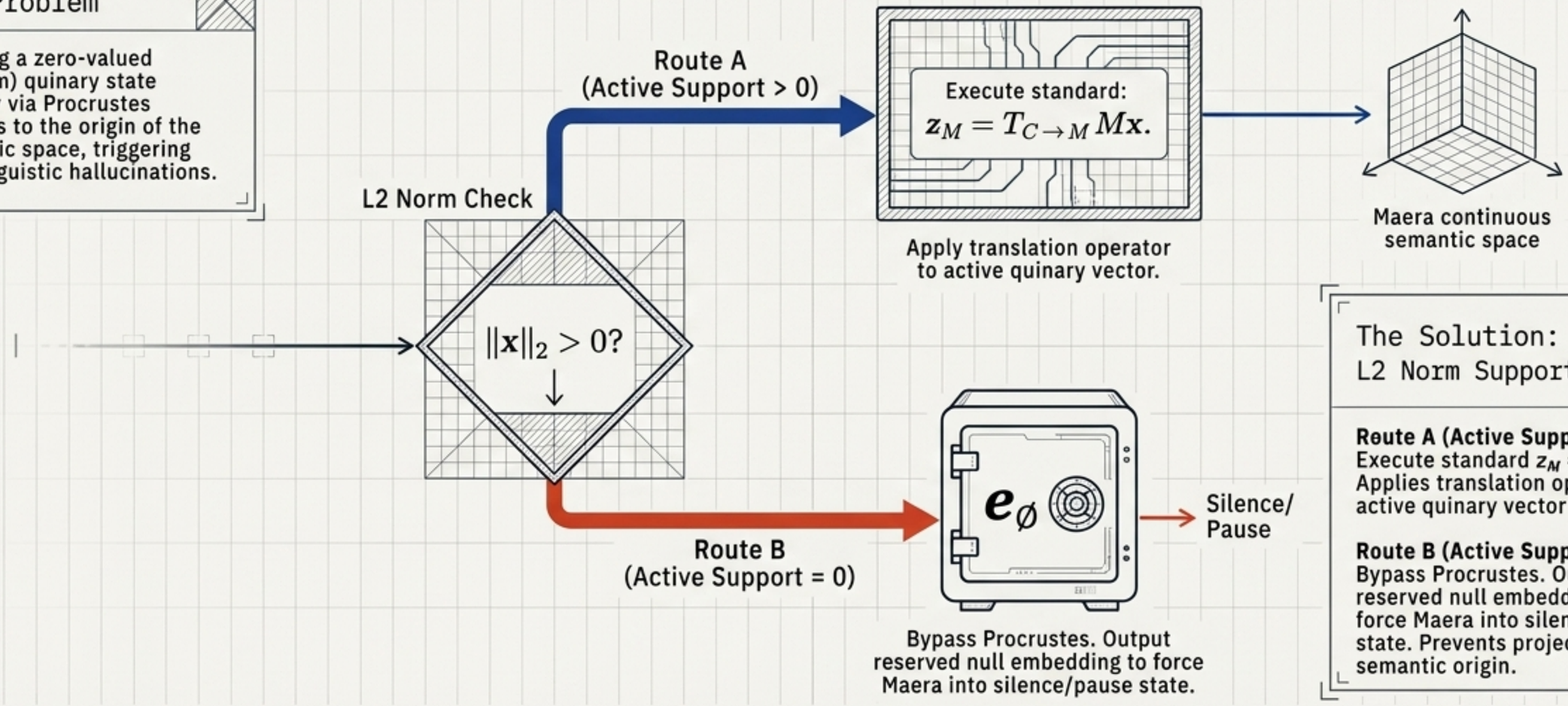
Factored operator processes ~92,000 multiply-adds.
Latency: < 2.0ms (Well below 50ms system

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Enforcing Semantic Integrity: Vacuum & Active-Mask Semantics

The Problem

Mapping a zero-valued (vacuum) quinary state directly via Procrustes projects to the origin of the semantic space, triggering LLM linguistic hallucinations.



The Solution: L2 Norm Support Routing

Route A (Active Support > 0): Execute standard $z_M = T_{C \to M} Mx$. Applies translation operator to active quinary vector for mapping.

Route B (Active Support = 0): Bypass Procrustes. Output reserved null embedding $e_\emptyset$ to force Maera into silence/pause state. Prevents projection to semantic origin.

SECURITY RULE: INACTIVE QUINARY COORDINATES MUST NEVER LEAK SIGN INFORMATION OR MANUFACTURE ARTIFICIAL SEMANTIC AGREEMENT.

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The Mandatory Honesty Clause: Boundaries of the Claim

What is Mathematically Proven

What is Mathematically Proven

Structural Isometry

Zero Gromov-Wasserstein distortion proves the two metric–measure spaces are isometric.
The protocol guarantees optimal relational correspondence.

It provides a stable, reproducible, automatic naming bijection without human labeling.

What is Not Proven

What is Not Proven

Symbol Grounding

The protocol does not solve Harnad's Symbol Grounding Problem.

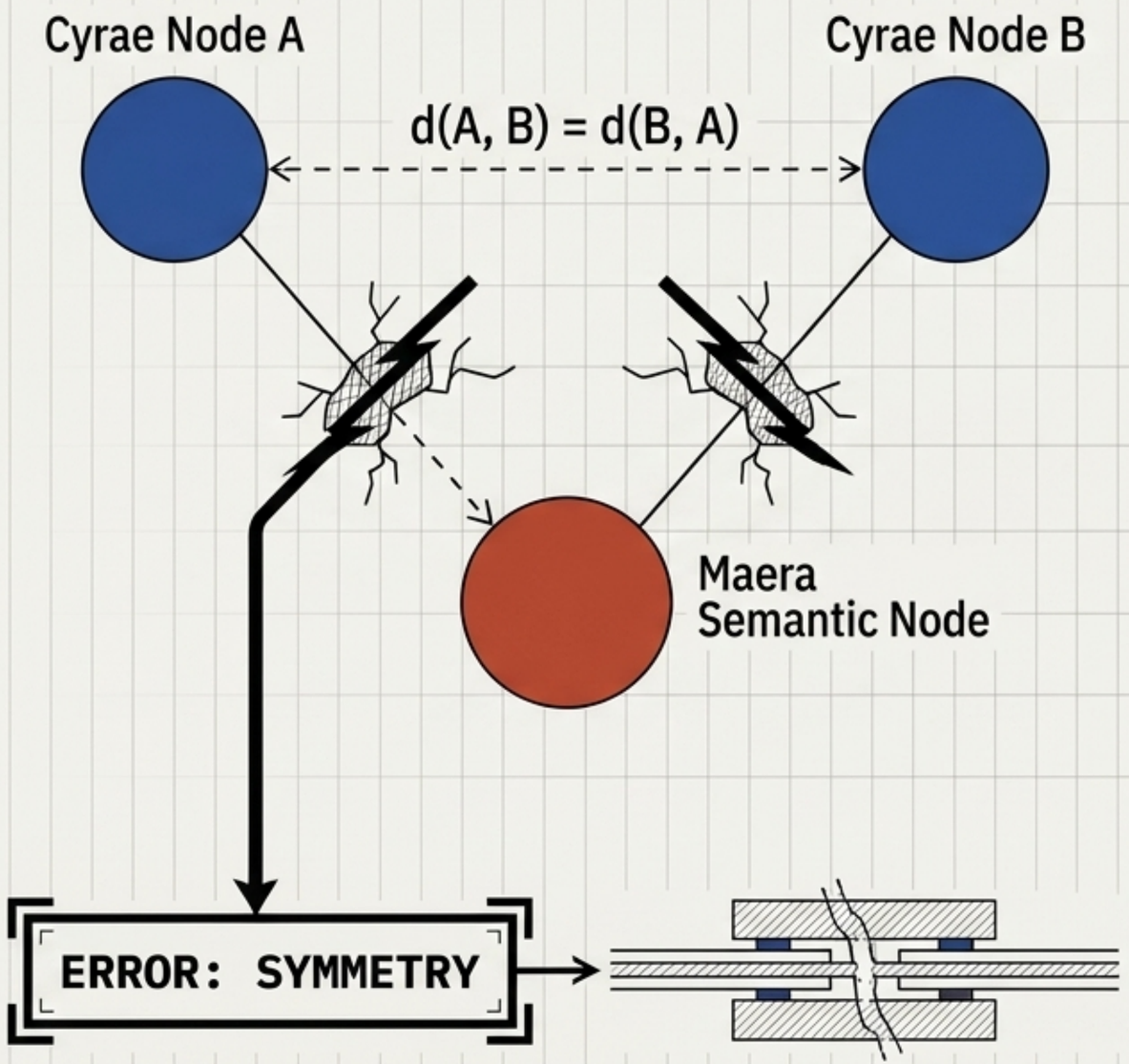
Cyrae's noise does not possess intrinsic English meaning.

Matching internal shape-graphs fixes which acoustic primitive is called "Threat"; it does not make the primitive mean threat.

CA/TUCR/HTLA canon must describe this as an "**automatic naming bijection**," never as "learning what words mean ex nihilo."

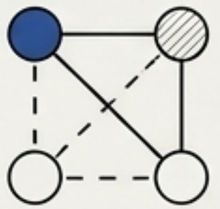
NotebookLM

The Identifiability Limit: Failing Closed on Symmetry



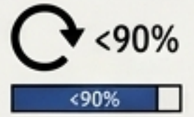
The Threat: NP-Hardness and Graph Automorphisms

If Cyrae's acoustic graph possesses structural symmetries (e.g., two indistinguishable primitive nodes), the quadratic assignment solution set is a full coset. Unsupervised recovery is mathematically impossible.

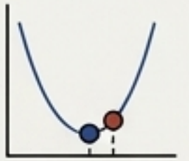


The Architecture Response: Strict "Fail Closed" Policy

Diagnostic Gate 1: Restart Stability (< 90% modal agreement across random restarts).



Diagnostic Gate 2: Assignment Margin (Low gap between first and second best assignments).



Action: If thresholds fail, emit **MAPPING_AMBIGUOUS** and enter quarantined bootstrap state.

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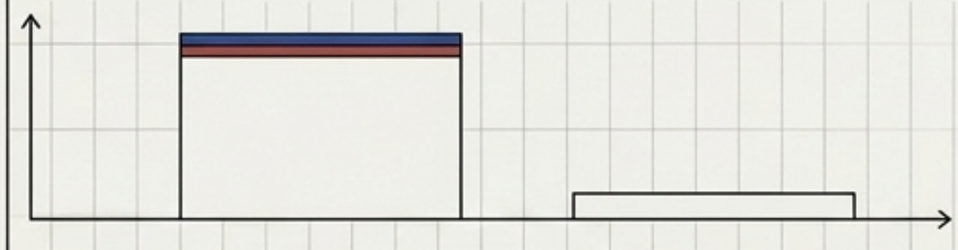
Validation Plan & Boot Sequence Diagnostics

Alignment Stress



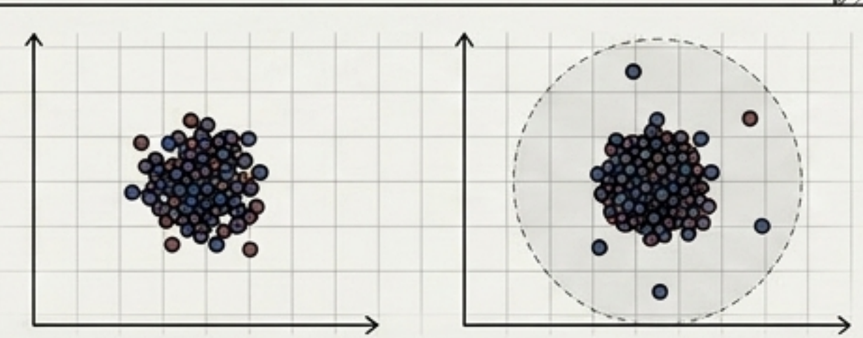
Normalized Frobenius stress S.
Measures residual geometry distortion.

Assignment Margin



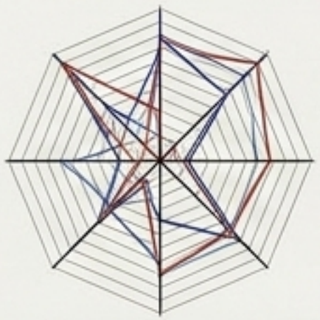
Gap between optimal and sub-optimal hard assignments. Measures identifiability.

Perturbation Stability



Application of controlled noise (ΔD_C) to the acoustic matrix. Measures fraction of assignment survival.

Seed-Set Symmetry Audit



Identifies near-symmetry in D_M. Flags structural interchangeability in Maera's seed concepts.

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Final Ratification & Bare-Metal Implementation

```
def fmp_gw_v1(C_a, C_s, p, q, lambda_reg):
    K = np.exp(-(C_a[:, np.newaxis, :] - C
    u = np.ones_like(p) / len(p)
    v = np.ones_like(q) / len(q)
```

NumPy Reference Code

The protocol is validated in a bare-metal, dependency-light NumPy implementation utilizing log-domain Sinkhorn stabilization and pure NumPy Hungarian solvers.

```
    for _ in range(max_iter):
        u = p / (K @ v)
        v = q / (K.T @ u)
    return np.diag(u) @ K @ np.diag(v)
```

Technical Manifest

```
mapping_protocol: FMP_GW_V1
acoustic_graph_hash: sha256:7a9c8b...
semantic_seed_hash: sha256:3d2f1e....
alignment_stress: 0.014
lock_scope: SPECIES_STABLE
```

The First Mapping Protocol is hereby ratified for Project Nexus v3.0 initialization.
Alignment is locked, persisted in manifest, and treated as permanent cognitive infrastructure.

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